11. Top-Down Parser for Context-Free Grammar

Aim: To implement a simple top-down parser for a context-free grammar using Python.

Procedure:

1. Define a context-free grammar.
2. Start parsing from the start symbol S.
3. Match grammar rules with the input sentence.
4. Check whether all words are consumed.
5. Display whether the sentence is accepted.

Code:

```
def parse(words):
    if len(words) == 3:
        if words[0] == "the" and words[1] == "boy" and words[2] == "runs":
            return True
    return False
sentence = "the boy runs"
words = sentence.split()
if parse(words):
    print("Sentence Accepted")
else:
    print("Sentence Rejected")
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/11.py"
Sentence Accepted
```

Result: A simple top-down parser was implemented to check a sentence against a grammar pattern.